

Tianmin Xie

Email: Tianmin.Xie@marshall.usc.edu $\diamond$ Website: tianminx.github.io

EDUCATION

- **University of Southern California** Aug 2022 - Present
Ph.D. in Data Sciences and Operations (Statistics) Advisor: Prof. Matteo Sesia
- **University of Cambridge** Oct 2021 - Jun 2022
Master of Mathematics (Part III)
Part III essay: Modern Changepoint Analysis (supervisor: Prof. Richard J. Samworth)
- **University of Cambridge** Oct 2018 - Jun 2021
B.A. in Mathematics
- **Tsinghua University** Sep 2018 - Oct 2018
Major: Mathematics (Transferred)
- **Shanghai High School** Sep 2015 - Jun 2018

RESEARCH INTERESTS

Machine Learning; Uncertainty Quantification; Conformal Inference; Deep Learning; Recommender Systems; Open-Set Recognition; High-dimensional Statistics; Causal Inference.

AWARDS

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| Marshall PhD Fellowship (\$10,000; competitive, 3 awardees/year) | May 2026 |
| Alexey Chervonenkis Best Poster Award (COPA 2025) | Sep 2025 |
| Magdalene College Scholar | Nov 2019 |
| Magdalene College Prize for excellence in University Examinations | Jun 2019 |
| Silver Medal in China Mathematical Olympiad (CMO) | Nov 2017 |
| First place (ranked #1), Chinese High School Mathematics League (Shanghai Region) | |
| (First Prize: 69 awardees; total awardees: 560) | Sep 2017 |

PUBLICATIONS AND PREPRINTS

Conformal Inference for Open-Set and Imbalanced Classification

T. Xie, Y. Zhou, Z. Liang, S. Favaro, M. Sesia.

<https://arxiv.org/abs/2510.13037>

Under major revision at *Biometrika* (2025)

Summary: Prediction sets with finite-sample coverage guarantees for classification when some classes are rare and unseen classes can appear at test time. Introduces conformal p-values for detecting new classes (with an optimality theory and a connection to the Good–Turing estimator) and a frequency-based selective data-splitting scheme that stays valid through re-weighting; validated on CelebA.

Structured Conformal Inference for Matrix Completion with Applications to Group Recommender Systems

Z. Liang*, T. Xie*, X. Tong, M. Sesia.

<https://arxiv.org/abs/2404.17561>

*The first two authors contributed equally to this work.

Journal of the American Statistical Association (2026)

Summary: A model-agnostic conformal wrapper for any matrix-completion algorithm that returns a joint confidence region for a group of missing entries (e.g., a household’s ratings of one movie). Achieves simultaneous group-level guarantees via a structured calibration set and a generalized weighted conformalization framework; validated on MovieLens 100K.

PRESENTATIONS

ASA Conference on Statistical Learning and Data Science (SLDS)	<i>Nov 2026 (upcoming)</i>
<i>Invited session</i>	<i>New York, NY</i>
Conformal and Probabilistic Prediction with Applications (COPA)	<i>Sep 2025</i>
<i>Presented on behalf of M. Sesia</i>	<i>London, United Kingdom</i>
Joint Statistical Meetings (JSM)	<i>Aug 2024</i>
<i>Topic-Contributed Paper Session</i>	<i>Portland, OR</i>
Causal Inference Seminar	<i>Fall 2023 - Spring 2024</i>
<i>Organized by Zijun Gao, Dennis Shen, Angela Zhou</i>	<i>University of Southern California</i>

TEACHING

BUAD 312: Statistics and Data Science for Business	<i>Fall 2025</i>
<i>Instructor</i>	<i>University of Southern California</i>
DSO Summer Scholars	<i>Summer 2025</i>
<i>Teaching Assistant</i>	<i>University of Southern California</i>
BUAD 312: Statistics and Data Science for Business	<i>Fall 2024</i>
<i>Teaching Assistant</i>	<i>University of Southern California</i>

PROFESSIONAL SERVICE

Conference Reviewer: NeurIPS 2026
Journal Referee: *Biometrika, Statistica Sinica*

SKILLS

Programming	Python, PyTorch, R, MATLAB, SQL, Git
Methods	Conformal inference, uncertainty quantification, distribution-free inference, statistical learning theory, statistical hypothesis testing, Bayesian inference, causal inference, robust statistics, stochastic calculus, convex optimization, machine learning (classification, regression, deep learning, imbalanced learning), functional data analysis, statistical modeling, time-series modeling, experimental design, matrix completion, collaborative filtering, recommender systems, cross-validation, missing data analysis, data science
Languages	English (fluent), Mandarin Chinese (native)

ACTIVITIES

Susquehanna Discovery Program	<i>Apr 2025</i>
<i>Susquehanna International Group</i>	<i>Bala Cynwyd, PA</i>
Princeton Machine Learning Theory Summer School	<i>Jun 2023</i>
<i>Organized by Boris Hanin (Princeton ORFE)</i>	<i>Princeton, NJ</i>
Causal Inference Reading Group	<i>Oct 2020 - Mar 2021</i>
<i>Organized by Qingyuan Zhao</i>	<i>University of Cambridge</i>
Semiparametric Statistics Reading Group	<i>Apr 2021 - Jul 2021</i>
<i>Organized by Rajen Shah and Qingyuan Zhao</i>	<i>University of Cambridge</i>

Last updated: October 5, 2026